Fri 14 Feb 2025

Last time, we ended with this problem:

Solve $\begin{pmatrix} 1\times10^{-4} & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

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exact arithmetic

$$\begin{pmatrix} 1 & 0 \\ 1\times10^{4} & 1 \end{pmatrix}\begin{pmatrix} 1\times10^{-4} & 1 \\ 0 & -9999 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Fwd solve $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -9998 \end{pmatrix}$ Back solve $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \frac{10000}{9999} \\ \frac{9998}{9999} \end{pmatrix}$

3digit decimal

$$\begin{pmatrix} 1 & 0 \\ 1.00\times10^{4} & 1 \end{pmatrix}\begin{pmatrix} 1.00\times10^{-4} & 1 \\ 0 & -1.00\times10^{4} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Fwd solve $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1.00 \\ -1.00\times10^{4} = fl\left(\frac{2-1\times10^{4}}{1}\right) \end{pmatrix}$

Back solve $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0.00 = fl\left(\frac{1.00-1.00}{1.00\times10^{-4}}\right) \\ 1.00 \end{pmatrix}$

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catastrophic cancellation!

We can trace the significant rounding error back to the elimination in the $\underline{b}$ vector where we had

$$2 - 1 * 1\times10^{4}$$

we lose this because this is so big

The multiplier was large because the value on the diagonal was so much smaller that the other values in the column.

An alternate interpretation:

When we factor $A = LU$

$$\underset{A}{\begin{pmatrix} 1.0\times10^{-4} & 0 \\ 1 & 1 \end{pmatrix}} = \underset{L}{\begin{pmatrix} 1 & 0 \\ 1\times10^{4} & 1 \end{pmatrix}} \underset{U}{\begin{pmatrix} 1\times10^{-4} & 1 \\ 0 & -9999 \end{pmatrix}}$$

we replace a problem $A\underline{x} = \underline{b}$ where

$\text{cond}(A) \doteq 4$ (small)

by two problems $L\underline{y} = \underline{b}$

$U\underline{x} = \underline{y}$

where $\text{cond}(L) \doteq 10^8$

$$L^{-1} = \begin{pmatrix} 1 & 0 \\ -1\times10^{4} & 1 \end{pmatrix}$$

$\text{cond}(U) \doteq 10^8$

$$U^{-1} = \begin{pmatrix} 1\times10^{4} & 10000/9999 \\ 0 & 1/9999 \end{pmatrix}$$

We have replaced a well conditioned problem by two ill-conditioned problems!

Fact Whenever multipliers are large (because diagonal elements are relatively small) a loss of accuracy may be observed.

Recall The exact algorithm fails when there is a 0 on the diagonal ... our solution was to re-order equations.

Here the floating-point algorithm failed and there was a near-0 on the diagonal
$\Longrightarrow$ Try reordering the equations

Try $R_1 \leftrightarrow R_2$

$$\begin{pmatrix} 1 & 1 \\ 1\times10^{-4} & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 \\ 1\times10^{-4} & 1 \end{pmatrix} \qquad U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

was 10^{+4} before

$\text{cond}(L) \doteq 1$
$\text{cond}(U) \doteq 4$
same cond # as original problem

Forward Solve $\begin{pmatrix} 1 & 0 \\ 1\times10^{-4} & 1 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $\quad y_1 = 2.00$, $y_2 = 1.00$

Backward Solve $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2.00 \\ 1.00 \end{pmatrix}$

$x_2 = 1.00$

The correctly rounded solution! $x_1 = 1.00$

∴ we should interchange rows using P_{i,r_i} to move small pivots off the diagonal during the LU factorization.

$$\begin{array}{c} \\ \\ \\ \\ \\ \end{array}\begin{array}{c} \begin{array}{ccccc} & & i & & \end{array} \\ \begin{pmatrix} \times & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times \\ 0 & 0 & \boxed{\times} & \times & \times \\ 0 & 0 & \boxed{\times} & \times & \times \\ 0 & 0 & \boxed{\times} & \times & \times \end{pmatrix} \end{array} \begin{array}{c} \\ \\ i \\ \\ r_i \end{array}$$

Choose interchange row r_i so that $|a_{r_i,i}| \geq |a_{j,i}|$ $\forall j$ in i to n

- this guarantees that all multipliers are ≤ 1 in size

∴ do row interchanges (aka partial pivoting) not just when the diagonal element is 0, but when it is smaller than elements that are to be eliminated.

What can we say about the accuracy of the computed solution?

Fact: In practice, when row interchanges are used, the Gaussian elimination algorithm is numerically stable, and generates approximate solutions $\underline{x}$ with relative residuals $\|\underline{r}\|/\|b\|$ of size $\approx C_n \cdot \varepsilon_{mach}$,

where C_n is a small constant that depends on n.

As a result,

$$\frac{\|\underline{x} - \underline{x}^*\|}{\|\underline{x}^*\|} \leq \text{Cond}(A) \cdot \frac{\|\underline{r}\|}{\|\underline{b}\|}$$

$$\sim \leq \text{Cond}(A) \cdot C_n \cdot \varepsilon_{mach}.$$

$\therefore$ the solution is accurate provided the problem is well-conditioned.

rule of thumb:

precision of your floating point arithmetic

$$\#\text{ bits of accuracy in computed sol}^{n} \approx p - \left\lceil \log_\beta \text{cond}(A) \right\rceil.$$

"In practice" means not always true. There are matrices for which Gaussian Elimination with Partial Pivoting produces inaccurate results for well conditioned problems.

The accuracy can be improved if <u>complete pivoting</u> is used.

In complete pivoting, the whole uneliminated submatrix is searched for the largest element

$$\begin{pmatrix} \times & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times \\ 0 & 0 & \times & \times & \times \\ 0 & 0 & \times & \times & \times \\ 0 & 0 & \times & \times & \times \end{pmatrix}$$

Suppose $|a_{s,t}| > |a_{i,j}|$ $\forall$ i,j,s,t in submatrix

Then we interchange rows and columns to move $a_{s,t}$ into the i,i pivot position.

This technique is called complete pivoting and the factorization becomes

$$PAQ = LU$$

We don't always use complete pivoting since it requires a 2-D search instead of a 1-D search

$$A\underline{x} = \underline{b}$$
$$PA\underline{x} = P\underline{b}$$
$$PAQQ^T\underline{x} = P\underline{b}$$
$$LU(Q^T\underline{x}) = P\underline{b}$$

Solve $L\underline{y} = P\underline{b}$

$U\underline{z} = \underline{y}$

$\underline{x} = Q\underline{z}$

Instead of interchanging columns (which corresponds to reordering unkown variables), use indirect addressing. Instead of refering to $a(i,j)$ refer to $a(rpiv(i), cpiv(j))$ where rpiv and cpiv are vectors that keep track of pivoting information